

Proof by Mathematical Induction

ANSWER KEY



The three steps of induction

- 1 State the three steps required for a valid proof by mathematical induction.
(1) Base case: verify the statement holds for the smallest value of n (usually $n = 1$). (2) Inductive hypothesis: assume the statement holds for $n = k$. (3) Inductive step: use the assumption to prove the statement holds for $n = k + 1$. Concluding, by the principle of mathematical induction, the statement holds for all n in the given range.

Proving a summation formula

- 2 Prove by induction that $1 + 2 + 3 + \dots + n = n(n + 1)/2$ for all $n \geq 1$.
Base case ($n = 1$): LHS = 1. RHS = $1(2)/2 = 1$. True. Inductive hypothesis: assume $1 + 2 + \dots + k = k(k + 1)/2$. For $n = k + 1$: $1 + 2 + \dots + k + (k + 1) = k(k + 1)/2 + (k + 1)$. Factoring out $(k + 1)$: this is $(k + 1)(k/2 + 1) = (k + 1)(k + 2)/2$, which matches the formula with $n = k + 1$. Since it holds for $n = 1$ and $n = k \Rightarrow n = k + 1$, by induction it holds for all $n \geq 1$.
- 3 Prove by induction that $1^2 + 2^2 + 3^2 + \dots + n^2 = n(n + 1)(2n + 1)/6$ for all $n \geq 1$.
Base case ($n = 1$): LHS = 1. RHS = $1(2)(3)/6 = 1$. True. Assume true for $n = k$:
 $1^2 + 2^2 + \dots + k^2 = k(k + 1)(2k + 1)/6$. For $n = k + 1$:
 $1^2 + \dots + k^2 + (k + 1)^2 = k(k + 1)(2k + 1)/6 + (k + 1)^2$. Factoring out $(k + 1)$: this is $(k + 1)[k(2k + 1)/6 + (k + 1)]$. Combining the bracketed terms over a denominator of 6: this is $(k + 1)(2k^2 + 7k + 6)/6$. Since $2k^2 + 7k + 6 = (2k + 3)(k + 2)$, this equals $(k + 1)(k + 2)(2k + 3)/6$, matching the formula at $n = k + 1$. By induction, true for all $n \geq 1$.

Proving divisibility results

- 4 Prove by induction that $7^n - 1$ is divisible by 6 for all $n \geq 1$.
Base case ($n = 1$): $7^1 - 1 = 6$, divisible by 6. Assume $7^k - 1 = 6m$ for some integer m , so $7^k = 6m + 1$. For $n = k + 1$: $7^{k+1} - 1 = 7 \cdot 7^k - 1 = 7(6m + 1) - 1 = 42m + 6 = 6(7m + 1)$, which is divisible by 6. By induction, $7^n - 1$ is divisible by 6 for all $n \geq 1$.

Induction with inequalities

- 5 Prove by induction that $2^n > n^2$ for all integers $n \geq 5$.
Base case ($n = 5$): $2^5 = 32$, $5^2 = 25$, and $32 > 25$. True. Assume $2^k > k^2$ for some $k \geq 5$. For $n = k + 1$: $2^{k+1} = 2 \cdot 2^k > 2k^2$. We need $2k^2 \geq (k + 1)^2 = k^2 + 2k + 1$, i.e. $k^2 - 2k - 1 \geq 0$, which holds for $k \geq 3$ (true since $k \geq 5$). So $2^{k+1} > 2k^2 \geq (k + 1)^2$. By induction, $2^n > n^2$ for all $n \geq 5$.

Induction in calculus: derivatives of a product

- 6 Prove by induction that $\frac{d^n}{dx^n}(e^{ax}) = a^n e^{ax}$ for all positive integers n .
Base case ($n = 1$): differentiating e^{ax} once gives $ae^{ax} = a^1 e^{ax}$, matching the formula. Inductive hypothesis: assume the k -th derivative of e^{ax} is $a^k e^{ax}$. Differentiating this once more:
 $\frac{d}{dx}(a^k e^{ax}) = a^k \cdot ae^{ax} = a^{k+1} e^{ax}$ — this is exactly the $(k + 1)$ -th derivative, matching the formula at $n = k + 1$. By induction, true for all positive integers n .
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Induction with a recursively defined sequence

- 7 A sequence is defined by $u_1 = 3$ and $u_{n+1} = 2u_n - 1$. Prove by induction that $u_n = 2^n + 1$ for all $n \geq 1$.
Base case ($n = 1$): $u_1 = 3$ (given). Formula: $2^1 + 1 = 3$. True. Assume $u_k = 2^k + 1$. Then
 $u_{k+1} = 2u_k - 1 = 2(2^k + 1) - 1 = 2^{k+1} + 2 - 1 = 2^{k+1} + 1$, matching the formula at $n = k + 1$. By induction, $u_n = 2^n + 1$ for all $n \geq 1$.
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Proving a geometric-series-style summation

- 8 Prove by induction that the sum $3^0 + 3^1 + \dots + 3^{n-1}$ equals $(3^n - 1)/2$ for all $n \geq 1$.
Base case ($n = 1$): LHS = $3^0 = 1$. RHS = $(3^1 - 1)/2 = 2/2 = 1$. True. Assume the sum $3^0 + \dots + 3^{k-1} = (3^k - 1)/2$. For $n = k + 1$: adding the next term 3^k gives $(3^k - 1)/2 + 3^k$. Combining over a denominator of 2: this is $(3^k - 1 + 2 \cdot 3^k)/2 = (3 \cdot 3^k - 1)/2 = (3^{k+1} - 1)/2$, matching the formula at $n = k + 1$. By induction, true for all $n \geq 1$.
- 9 Prove by induction that the sum $1 \cdot 1! + 2 \cdot 2! + \dots + n \cdot n!$ equals $(n + 1)! - 1$ for all $n \geq 1$.
Base case ($n = 1$): LHS = $1 \cdot 1! = 1$. RHS = $2! - 1 = 2 - 1 = 1$. True. Assume $1 \cdot 1! + \dots + k \cdot k! = (k + 1)! - 1$. For $n = k + 1$: adding the next term $(k + 1)(k + 1)!$ gives $(k + 1)! - 1 + (k + 1)(k + 1)! = (k + 1)! [1 + (k + 1)] - 1 = (k + 1)! (k + 2) - 1 = (k + 2)! - 1$, matching the formula at $n = k + 1$. By induction, true for all $n \geq 1$.
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Proving an inequality involving factorials

- 10 Prove by induction that $n! > 2^n$ for all integers $n \geq 4$.
Base case ($n = 4$): $4! = 24$, $2^4 = 16$, and $24 > 16$. True. Assume $k! > 2^k$ for some $k \geq 4$. For $n = k + 1$: $(k + 1)! = (k + 1) \cdot k! > (k + 1) \cdot 2^k$. Since $k \geq 4$, $k + 1 \geq 5 > 2$, so $(k + 1) \cdot 2^k > 2 \cdot 2^k = 2^{k+1}$. Combining, $(k + 1)! > 2^{k+1}$, matching the inequality at $n = k + 1$. By induction, $n! > 2^n$ for all $n \geq 4$.
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Proving a divisibility result with two variables

- 11** Prove by induction that $n^3 + 2n$ is divisible by 3 for all positive integers $n \geq 1$.

Base case ($n = 1$): $1^3 + 2(1) = 3$, divisible by 3. Assume $k^3 + 2k = 3m$ for some integer m . For $n = k + 1$: $(k + 1)^3 + 2(k + 1) = k^3 + 3k^2 + 3k + 1 + 2k + 2$. Grouping terms: this is $(k^3 + 2k) + (3k^2 + 3k + 3) = 3m + 3(k^2 + k + 1)$. Factoring out 3: $3m + 3(k^2 + k + 1) = 3(m + k^2 + k + 1)$, which is divisible by 3. By induction, $n^3 + 2n$ is divisible by 3 for all $n \geq 1$.

- 12** Prove by induction that $9^n - 1$ is divisible by 8 for all positive integers $n \geq 1$, and explain briefly why a similar argument would show $a^n - 1$ is divisible by $a - 1$ for any integer $a > 1$.

Base case ($n = 1$): $9^1 - 1 = 8$, divisible by 8. Assume $9^k - 1 = 8m$ for some integer m , so $9^k = 8m + 1$. For $n = k + 1$: $9^{k+1} - 1 = 9 \cdot 9^k - 1 = 9(8m + 1) - 1 = 72m + 9 - 1 = 72m + 8 = 8(9m + 1)$, divisible by 8. By induction, true for all $n \geq 1$. The same inductive pattern works for any base a : $a^{k+1} - 1 = a \cdot a^k - 1 = a[(a - 1)m' + 1] - 1 = (a - 1)(am') + (a - 1) = (a - 1)(am' + 1)$, always a multiple of $a - 1$.

Induction with a recurrence relation for a sequence of sums

- 13** A sequence satisfies $S_1 = 2$ and $S_{n+1} = S_n + 6n + 2$ for $n \geq 1$. Prove by induction that $S_n = 3n^2 - n$ for all $n \geq 1$.

Base case ($n = 1$): $S_1 = 2$ (given). Formula: $3(1)^2 - 1 = 2$. True. Assume $S_k = 3k^2 - k$. Then $S_{k+1} = S_k + 6k + 2 = 3k^2 - k + 6k + 2 = 3k^2 + 5k + 2$. The formula at $n = k + 1$ gives $3(k + 1)^2 - (k + 1) = 3k^2 + 6k + 3 - k - 1 = 3k^2 + 5k + 2$, which matches exactly. Since it holds for $n = 1$ and $n = k \Rightarrow n = k + 1$, by induction $S_n = 3n^2 - n$ for all $n \geq 1$.

Induction with a strengthened hypothesis

- 14** Prove by induction that $11^n - 4^n$ is divisible by 7 for all positive integers $n \geq 1$.

Base case ($n = 1$): $11^1 - 4^1 = 7$, divisible by 7. Assume $11^k - 4^k = 7m$ for some integer m , so $11^k = 7m + 4^k$. For $n = k + 1$: $11^{k+1} - 4^{k+1} = 11 \cdot 11^k - 4 \cdot 4^k$. Substituting $11^k = 7m + 4^k$: this is $11(7m + 4^k) - 4 \cdot 4^k = 77m + 11 \cdot 4^k - 4 \cdot 4^k$. Combining the 4^k terms: $77m + 7 \cdot 4^k$. Factoring out 7: $7(11m + 4^k)$, which is divisible by 7. By induction, $11^n - 4^n$ is divisible by 7 for all $n \geq 1$.

Induction to prove a formula for the sum of cubes

- 15** Prove by induction that $1^3 + 2^3 + 3^3 + \dots + n^3 = [n(n + 1)/2]^2$ for all $n \geq 1$.

Base case ($n = 1$): LHS = $1^3 = 1$. RHS = $[1(2)/2]^2 = 1^2 = 1$. True. Assume $1^3 + \dots + k^3 = [k(k + 1)/2]^2$. For $n = k + 1$: adding $(k + 1)^3$ gives $[k(k + 1)/2]^2 + (k + 1)^3$. Factoring out $(k + 1)^2$: this is $(k + 1)^2[k^2/4 + (k + 1)]$. Combining over a denominator of 4: this is $(k + 1)^2(k^2 + 4k + 4)/4$. Since $k^2 + 4k + 4 = (k + 2)^2$, this is $(k + 1)^2(k + 2)^2/4 = [(k + 1)(k + 2)/2]^2$, matching the formula at $n = k + 1$. By induction, true for all $n \geq 1$.

- 16** Prove by induction that $5^n + 2 \cdot 3^{n-1} + 1$ is divisible by 8 for all positive integers $n \geq 1$.
 Base case ($n = 1$): $5^1 + 2(3^0) + 1 = 5 + 2 + 1 = 8$, divisible by 8. Assume $5^k + 2 \cdot 3^{k-1} + 1 = 8m$ for some integer m . Consider the difference between consecutive terms:
 $(5^{k+1} + 2 \cdot 3^k + 1) - (5^k + 2 \cdot 3^{k-1} + 1) = 5^{k+1} - 5^k + 2 \cdot 3^k - 2 \cdot 3^{k-1}$. This simplifies to
 $4 \cdot 5^k + 4 \cdot 3^{k-1} = 4(5^k + 3^{k-1})$. Since 5^k and 3^{k-1} are both odd, their sum is even, so $4(5^k + 3^{k-1})$ is a multiple of 8. Therefore $5^{k+1} + 2 \cdot 3^k + 1 = 8m + 4(5^k + 3^{k-1})$, and since both terms are multiples of 8, so is their sum. By induction, the statement holds for all $n \geq 1$.

Induction to prove a formula for a weighted geometric sum

- 17** Prove by induction that $1 \cdot 2^1 + 2 \cdot 2^2 + 3 \cdot 2^3 + \dots + n \cdot 2^n = (n-1)2^{n+1} + 2$ for all $n \geq 1$.
 Base case ($n = 1$): LHS = $1 \cdot 2^1 = 2$. RHS = $(1-1)2^2 + 2 = 0 + 2 = 2$. True. Assume
 $1 \cdot 2^1 + \dots + k \cdot 2^k = (k-1)2^{k+1} + 2$. For $n = k+1$: adding the next term $(k+1)2^{k+1}$ gives
 $(k-1)2^{k+1} + 2 + (k+1)2^{k+1}$. Combining the 2^{k+1} terms: this is
 $[(k-1) + (k+1)]2^{k+1} + 2 = 2k \cdot 2^{k+1} + 2 = k \cdot 2^{k+2} + 2$. The formula at $n = k+1$ requires
 $[(k+1)-1]2^{k+2} + 2 = k \cdot 2^{k+2} + 2$, which matches exactly. By induction, the formula holds for all
 $n \geq 1$.

A closing check: identifying an invalid induction argument

- 18** A student claims to prove that all horses are the same color using induction on the number of horses n in a group: the base case $n = 1$ is trivially true (one horse is trivially "the same color" as itself), and the student then argues that if any k horses are the same color, then any $k+1$ horses must also be the same color. Explain precisely where this argument breaks down, and state why a correct induction proof must avoid this flaw.

The flaw is in the inductive step: the argument tries to overlap two groups of k horses (the first k and the last k out of $k+1$ total) and conclude they share a common color through the overlap. This overlap argument only works when the overlap is non-empty, which requires $k \geq 2$. It fails silently at the transition from $n = 1$ to $n = 2$, since two groups of 1 horse each (drawn from a total of 2) have no horse in common, so there is no shared reference point linking their colors. This illustrates why a correct induction proof must verify that the inductive step logic genuinely holds for the smallest values of k used in the argument, not just assume it holds uniformly for all k .

Proving a sum-of-products formula by induction

- 19** Prove by induction that $1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \dots + n(n+1) = n(n+1)(n+2)/3$ for all $n \geq 1$.
 Base case $n = 1$: LHS = $1 \cdot 2 = 2$, RHS = $1(2)(3)/3 = 2$. Equal, so the base case holds. Inductive step: assume $1 \cdot 2 + \dots + m(m+1) = m(m+1)(m+2)/3$ for some $m \geq 1$. Adding $(m+1)(m+2)$ to both sides: the new sum is $m(m+1)(m+2)/3 + (m+1)(m+2)$. Factoring out $(m+1)(m+2)$: this is
 $(m+1)(m+2)[m/3 + 1] = (m+1)(m+2)(m+3)/3$ — exactly the claimed formula with $n = m+1$.
 By induction, the formula holds for all $n \geq 1$.

Proving an exponential inequality by induction

20 Prove by induction that $2^n > n$ for all integers $n \geq 1$.

Base case $n = 1$: $2^1 = 2 > 1$, true. Inductive step: assume $2^m > m$ for some $m \geq 1$. Then $2^{m+1} = 2 \cdot 2^m > 2m$ (using the assumption). Since $m \geq 1$, $2m \geq m + 1$, so $2^{m+1} > 2m \geq m + 1$. Thus $2^{m+1} > m + 1$. By induction, $2^n > n$ for all $n \geq 1$.

Proving a divisibility result by induction

21 Prove by induction that $n^3 - n$ is divisible by 6 for all integers $n \geq 1$.

Base case $n = 1$: $1^3 - 1 = 0$, and 0 is divisible by 6. Inductive step: assume $m^3 - m = 6k$ for some integer k . Then $(m+1)^3 - (m+1) = m^3 + 3m^2 + 3m + 1 - m - 1$. Simplifying: this is $m^3 + 3m^2 + 2m$. Regrouping: this is $(m^3 - m) + 3m(m+1) = 6k + 3m(m+1)$. Since m and $m+1$ are consecutive integers, one of them is even, so $m(m+1)$ is even and $3m(m+1)$ is divisible by 6. Thus $(m+1)^3 - (m+1) = 6k + 6j = 6(k+j)$ for some integer j , which is divisible by 6. By induction, $n^3 - n$ is divisible by 6 for all $n \geq 1$.

Proving a closed form for a recursive sequence

22 A sequence is defined by $a_1 = 1$ and $a_{n+1} = 2a_n + 1$ for $n \geq 1$. Prove by induction that $a_n = 2^n - 1$ for all $n \geq 1$.

Base case $n = 1$: $a_1 = 1$, and $2^1 - 1 = 1$. Equal, so the base case holds. Inductive step: assume $a_m = 2^m - 1$ for some $m \geq 1$. Then $a_{m+1} = 2a_m + 1 = 2(2^m - 1) + 1 = 2^{m+1} - 2 + 1 = 2^{m+1} - 1$ — exactly the claimed formula with $n = m + 1$. By induction, $a_n = 2^n - 1$ for all $n \geq 1$.

A second exponential inequality by induction

23 Prove by induction that $3^n \geq 2n + 1$ for all integers $n \geq 1$.

Base case $n = 1$: $3^1 = 3$ and $2(1) + 1 = 3$, so $3 \geq 3$ holds (with equality). Inductive step: assume $3^m \geq 2m + 1$ for some $m \geq 1$. Then $3^{m+1} = 3 \cdot 3^m \geq 3(2m + 1) = 6m + 3$. Since $m \geq 1$, $4m \geq 0$, so $6m + 3 \geq 2m + 3 = 2(m + 1) + 1$. Thus $3^{m+1} \geq 2(m + 1) + 1$. By induction, $3^n \geq 2n + 1$ for all $n \geq 1$.

A third inequality by induction

- 24** Prove by induction that $n^2 \geq 2n - 1$ for all integers $n \geq 1$.

Base case $n = 1$: $1^2 = 1$ and $2(1) - 1 = 1$, so $1 \geq 1$ holds (with equality). Inductive step: assume $k^2 \geq 2k - 1$ for some $k \geq 1$. Then $(k + 1)^2 = k^2 + 2k + 1 \geq (2k - 1) + 2k + 1 = 4k$. Since $k \geq 1$, $2k \geq 1$, so $4k \geq 2k + 1 = 2(k + 1) - 1$. Thus $(k + 1)^2 \geq 2(k + 1) - 1$. By induction, $n^2 \geq 2n - 1$ for all $n \geq 1$.

Induction in combinatorics: counting subsets

- 25** Prove by induction that a set with n elements has exactly 2^n subsets, for all integers $n \geq 1$.

Base case $n = 1$: a 1-element set $\{a\}$ has exactly 2 subsets ($\emptyset$ and $\{a\}$), and $2^1 = 2$. True. Inductive step: assume every k -element set has 2^k subsets. Consider a $(k + 1)$ -element set formed by adding one new element x to a k -element set. Every subset of the $(k + 1)$ -element set either excludes x (giving exactly the 2^k subsets of the original k -element set) or includes x (giving one more subset for each of those same 2^k subsets, with x added in). This gives $2^k + 2^k = 2 \cdot 2^k = 2^{k+1}$ subsets in total. By induction, an n -element set has 2^n subsets for all $n \geq 1$.

Proving the sum of the first n odd numbers

- 26** Prove by induction that $1 + 3 + 5 + \dots + (2n - 1) = n^2$ for all $n \geq 1$.

Base case $n = 1$: LHS = 1, RHS = $1^2 = 1$. True. Inductive step: assume $1 + 3 + \dots + (2k - 1) = k^2$ for some $k \geq 1$. Adding the next odd number, $2(k + 1) - 1 = 2k + 1$, gives $k^2 + 2k + 1$. Since $k^2 + 2k + 1 = (k + 1)^2$, this matches the formula with $n = k + 1$. By induction, $1 + 3 + \dots + (2n - 1) = n^2$ for all $n \geq 1$.